1 A twelve-positive-variable polynomial of degree 901

The inner radical-separating multiplier can be removed, rather than merely made cheaper. The resulting single equation has exact degree 901 and 112 binary arithmetic operations. Its three-equation presentation has maximum degree 892 and 107 operations, or 106 after the inherited positive transport shift. A shorter carrier gives degree 905 in 110 operations (shifted system: degree 896, 104 operations). Every binary $+, -, \times$, including multiplication by a numeral, is charged; comparisons in systems are free. The degree-901 form improves the pinned 1022/124 form in both degree and cost. These are ordinary mathematical reductions formalizable in ZFC, not consistency proofs or new proof-assistant certificates.

Fixed instance and explicit definition. Select the complete source31 Wang943 121-by-8801 tuple: $\beta = 2^{1064921} - 1$, $K_0 = K$, $M_C = M_C$, $M_F = M_{F,\mathrm{source}}$, with the *shifted* mask. The included coefficient binding gives the immutable source adapter and all four materialized payload hashes. The twelve positive unknowns are $J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t$; all following letters are computed polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & X &= wq, & Y &= sq^3, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, & H &= 4a + 3,\\
\Delta &= a^2 + H, & \ell &= EkY, & A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2,\\
b_0 &= X + ac, & I &= ic^2, & D &= 1 + \Delta I^2, & Q &= (\Delta I)^2,\\
d &= Q - 1, & v &= cT, & b &= c + RD, & e &= Q(Dv^2 + b^2) - 1,\\
B_* &= 2Qvb, & K_* &= IT + 1, & L &= 1 + HK_*, & W &= b_0 + H(t + b + \delta),
\end{aligned}$$

where

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_k &= k - hE - R, \qquad N_t = (K_0 + w)(q - F - \alpha) + q - F - z(q-1),\\
\delta &= (a+2)c - \ell \quad \text{(full carrier)}, \qquad \delta = ac - \ell \quad \text{(short carrier)}.
\end{aligned}$$

The integer polynomial N is the eight-factor conjugate product

$$N = \prod_{g=\pm\sqrt{G},\, \tau=\pm\sqrt{A},\, f=\pm\sqrt{D}} \left[d(W - Lg - H\tau)^2 - H^2 e + H^2 B_* f\right].$$

Formal radical signs specify a polynomial, not extra unknowns or independent Galois automorphisms. The equation and system are

$$N_k N_t (1 + N) - 1 = 0, \qquad \text{or} \qquad N_k = 1, \quad N_t = 1, \quad N = 0.$$

For an explicit division-free evaluator put

$$\begin{aligned}
u &= dH^2 A, & p &= H^2 QDv^2, & r &= H^2 Qb^2, & j &= 4pr,\\
e_H &= p + r - H^2, & \mathcal{T} &= 2d(W^2 + L^2 G), & \mathcal{K} &= [d(W^2 - L^2 G)]^2,\\
a_* &= \mathcal{T} - 2(u + e_H), & b_* &= (u - e_H)^2 - j - \mathcal{K},\\
v_* &= 16uj - a_*(a_*\mathcal{T} + 2b_*), & u_* &= b_*^2 - a_*^2 \mathcal{K},\\
N &= \mathcal{K} v_*^2 - u_*(\mathcal{T} v_* - u_*).
\end{aligned}$$

Eliminating τ, f gives $(x^2 - 2(u + e_H)x + (u - e_H)^2 - j)^2 - 16ujx$; reducing modulo $x^2 - \mathcal{T}x + \mathcal{K}$ proves the formula. Also $N \equiv (u_* - d(W^2 + L^2 G)v_*)^2 \pmod 4$. Hence a product root forces $N = 0$ and $N_k = N_t = \epsilon \in \{1, -1\}$.

The two new recovery lemmas. The unchanged weak compiler interface gives

$$q \geq 16, \quad Y \geq q^3, \quad E = XY \geq q^4, \quad 0 < R < E, \quad k > E, \quad kY < c < 2kY, \quad a + 2 < 2E.$$

For estimates let $g = \sqrt{G}, f = \sqrt{D}, \tau = \sqrt{A}$ be positive real roots. Direct expansion gives $1 < A < \Delta c^2 \leq \Delta I = \sqrt{Q}$ and $g > \tau$. Moreover $g < 2Ec$, $f < 2EI$, $I \geq c^2$ and $c > EY$ imply

$$\begin{aligned}
b + cf &< 5E^3 I^2 + 2EI^2 < 6E^3 I^2 < \tfrac{1}{2} E^4 I^2 < d,\\
Ig - cf &= \frac{I^2 - c^2}{Ig + cf} > \frac{I}{8Ec} = \frac{ic}{8E} > \frac{Y}{8} > 1.
\end{aligned}$$

The second line uses the exact identity $(Ig)^2-(cf)^2=I^2-c^2$. For

$$C(x)=d(x^2+A)-e,\quad F(x)=[C(x)^2+DB_*^2-4d^2x^2A]^2-4C(x)^2DB_*^2,$$

every complex root has the coherent form $x=\tau'+y$, $(\tau')^2=A$, $(f')^2=D$, $dy^2=Q(vf'-b)^2-1$. Writing $Z=b+vf>1$ gives

$$|y|<Z(1+1/d),\qquad Z/d<T,\qquad |y|<b+ITg,\qquad |x|<b+(IT+1)g.$$

Here the first bound follows by squaring, the second from $b+cf<d$, and the third from $T(Ig-cf)>T$. At a norm zero, a negative main root would instead give

$$x=(W+Lg)/H=b+t+\delta+b_0/H+(K_*+1/H)g>b+K_*g,$$

a contradiction. Irrational quadratic conjugation would force that negative branch as well. Thus the selected g is a positive integer and $x=(W-Lg)/H$ is rational. This argument covers imaginary auxiliary roots.

The crucial inner recovery no longer needs a large multiplier. If $C(x)=0$ at a root of F, substitution of $x^2=e/d-A$ and $B_*=2Qvb$, followed by exact division by 4, gives

$$d^2A^2-deA+Q^2v^2b^2D=0.$$

Reducing modulo Q forces $Q\mid A(A-1)$, impossible because $0<A(A-1)<Q$. Likewise $F(0)=0$ forces $Q\mid(A-1)^2$, also impossible. Consequently the justified rational formulas

$$f'=-\frac{C(x)^2+DB_*^2-4d^2x^2A}{2C(x)B_*},\qquad \tau'=\frac{C(x)+B_*f'}{2dx},\qquad y=x-\tau'$$

recover integer square roots f',τ'. The inherited rational Pell saturation lemma applied to $Q=(\Delta I)^2$ makes y integral. Reduction modulo H gives $\gamma=(g-b_0)/H\in\mathbb{Z}$, and $g>(a+1)c$, $c>X$ give $\gamma>0$. The normalized signed-quotient compiler interface now reconstructs the accepting blank K215 computation.

Completeness and positivity are constructive. For a canonical positive compiler witness, $V=vf-b$ is a nonzero integer, so $y^2=V^2+(V^2-1)/d\geq V^2$ and $y\geq b-vf$. Both corrections satisfy $0<\delta<g$, since $\ell/c>E/2>2$. Set

$$t=\gamma+(IT+1)g+\tau+y-b-\delta>\gamma+\tau+ITg-vf>0.$$

Then $W=Lg+H(\tau+y)$, giving $N=0$ and $N_k=N_t=1$. No new Pell congruence, enlarged index, or divisibility restriction is used. The inherited substitution $z_{\mathrm{old}}=z+1$ remains positive after compiler reconstruction and removes one system gate. Composing with the existing K215 calibration and negating proves in ZFC that absence of positive solutions is equivalent to Con(ZFC).

Exact degree and verification. Scale all twelve unknowns by a formal parameter. Put $q_0=\beta J$, $k_0=\eta+\zeta$, $Y_0=sq_0^3$, $E_0=wq_0Y_0$, $c_0=k_0Y_0$, $I_0=ic_0^2$, $H_0=4E_0$, $\ell_0=E_0c_0$, $d_0=E_0^4I_0^2$, $B=q_0^3(q_0-F)E_0^2I_0^2$, and $V_0=I_0T\ell_0$. Four conjugate factors have degree 119; cancellation of the matching f,g signs leaves degrees $107,107,105,97$ for the full carrier. Their product has nonzero leading form

$$-2^{14}H_0^{16}d_0^8B^8V_0^4\ell_0^2k_0Y_0^2\left(2t+k_0-\frac{iTc_0}{E_0}\right),$$

of degree 892. The apparent denominator cancels against the prefactors; independent occurrence of t proves nonvanishing for every fixed $\beta>0$. For the short carrier the final factor instead has degree 101 and the leading form is $2^{16}H_0^{16}d_0^8B^8V_0^4\ell_0^2k_0Y_0^2c_0$, of degree 896. The nonzero leading forms of N_k,N_t add $7+2$ degrees. The full proof supplies every cancellation and denominator check. The reproducible checker independently multiplies the formal radical factors, checks signed and degenerate fixtures, and verifies these complete polynomial ray degrees and analytic leading coefficients over three finite fields. Finite diagnostics do not substitute for the unbounded compiler theorem.